

BrineX - Research Maturity Report

1. Clear Evaluation Metrics Defined

A. Crystal Module (BrineX-CF - Crystallization Forecasting)

LSTM - Daily Parameter Prediction:

Metric	Value	Description
Validation R ²	0.8884	88.84% variance explained on validation set
Test R ²	0.7750	77.50% accuracy on unseen test data
Test MAE	0.2264	Mean Absolute Error on test set
Test RMSE	0.3651	Root Mean Squared Error on test set
Input Features	15	8 crystallization parameters + 7 weather features
Input Window	30 days	Sliding window per inference
Output	480 values	60 days × 8 parameters per inference

Linear Regression - Monthly Production Calibration:

Metric	Description
Method	Converts LSTM daily parameter predictions to monthly production volumes
Alternatives Considered	ARIMA, Holt-Winters - rejected because they model historical patterns only, not causal inputs
Selected Model	Linear Regression - models causal inputs (beds, rain, temperature, seasonality) directly
R ²	0.9732 - empirically confirms causal model outperforms autoregressive alternatives
Confidence Interval	95% prediction intervals derived from actual regression residuals (1.96 × residual std)
Output Scope	Seasonal (Maha/Yala), monthly totals, landowner-wise scaled by bed count

B. Compass Module (BrineX-IPE - Intelligent Planning)

Demand Forecasting - Yield Ratio Algorithm:

Metric	Value	Description
Method	Seasonal yield ratio × production forecast	Per-landowner demand estimation
Yala Season Ratio	8.74%	% of production converting to immediate deals
Maha Season Ratio	10.27%	Higher due to distributor quality preference
Training Data	168 months PSS aggregate	2012–2025 government dataset
Validation Error	~10% MAPE	Validated against held-out PSS historical data

Input	Crystal production forecast (landowner-scaled)	Bed count ratio already applied by Crystal
Future Improvement	Farm-specific ratios after 6–12 months of deal history	Automatic transition — no code changes needed

Price Forecasting - SARIMAX Model:

Metric	Value	Description
Model Specification	SARIMAX(1,1,1)(0,1,1,12)	Seasonal time-series model
MAPE Month+1	1.16%	Out-of-sample validated
MAPE Month+2	1.35%	Two-pass chained prediction
Directional Accuracy	46.3%	Low - expected due to mean-reverting market behavior
Runtime Inputs	4	Season flag, price lag 12m, latest price, PSS production volume
Training Data	168 months PSS price history	2012–2025
Forecasting Method	Two-pass recursive	Month+2 uses month+1 prediction as input

Seller Recommendation:

- No ML model - sellers ranked transparently by highest price per bag and total bag count
- Fully auditable, no hidden algorithm bias

C. Vision Module (BrineX-QC - Computer Vision Quality Control)

Metric	Value	Description
Processing Time	~45ms/frame	End-to-end inference latency
Detection Classes	3	Pure, Impure, Unwanted
Confidence Threshold	0.50	Minimum detection confidence
IoU Threshold	0.45	Non-Max Suppression overlap threshold
Input Resolution	320×320	Resized via Lanczos3 + CHW tensor normalization
Throughput	~20+ FPS	CPU-optimized ONNX session
Purity Calculation	$(\text{pureCount} / (\text{pureCount} + \text{impureCount})) \times 100$	Per frame, per batch, per session
Whiteness Score	$V \times (1 - S) \times 100$	HSV-based per-pixel color analysis
Quality Score	×1.0 pure, ×0.3 impure, ×0 unwanted	Aggregated per crystal, per batch

D. Valor Module (BrineX-WV - Federated Waste Valorization)**Ensemble Model - Waste Prediction:**

Metric	Value	Description
Overall R ²	0.98125	~98% variance explained on held-out test data
MAE	~3,999.83	Mean Absolute Error across all targets
RMSE	~9,484.82	Root Mean Squared Error across all targets
Per-target R ²	0.9676 – 0.9907	Strong consistency across all 8 waste categories
Inference Time	~285ms	Mean single-row prediction, measured across 50 runs

Output Categories:

Type	Outputs
Solid Waste (kg)	Total waste, Limestone, Gypsum, Industrial Salt
Liquid Waste (L)	Bittern
Valorization Products	Magnesium Oil (L), Epsom Salt (kg), Potash (kg)

Model Inputs:

- Primary: production volume, rain sum, temperature mean, humidity mean, wind speed mean, month
- Engineered: log/scaled transformations, production × rainfall interactions, sin/cos seasonal encoding, evaporation index, brine density proxy

2. Baseline Comparison

Component	Baseline (Before)	BrineX Implementation	Improvement
Salt Quality Assessment	Manual visual inspection - subjective, slow, inconsistent	Automated YOLOv8 3-class detection + HSV whiteness analysis	Objective, repeatable, real-time (~45ms/frame)
Daily Parameter Forecasting	Manual observation, experience-based	LSTM model - 88.84% validation accuracy, 60-day rolling forecast	Data-driven daily operational planning
Monthly Production Forecasting	No prediction capability	Linear Regression calibration from LSTM output - ARIMA tried and rejected	Seasonal and landowner-wise forecasts with ±15% confidence
Production Monitoring	Paper-based recording, no weather integration	Digital daily measurement capture with OpenWeatherMap integration	Automated weather enrichment, historical trend visualization

Demand Forecasting	No market visibility	Yield ratio algorithm - production forecast $\times$ seasonal ratio (Yala: 8.74%, Maha: 10.27%)	-landowner demand estimates linked directly to Crystal production forecasts - ~10% MAPE on held-out PSS data
Price Forecasting	Phone-based negotiation, no market data	SARIMAX price model — MAPE 1.16% month+1, 1.35% month+2, two-pass recursive forecasting	Market price predictions capturing Maha/Yala seasonal structure and supply volume sensitivity
Seller Discovery	Manual phone calls, no transparency	Ranked by highest price and bag count - fully auditable	Transparent, fair market discovery for landowners
Waste Management	Manual estimation, operator guesswork, fixed averages - ~60–65% accuracy	Ensemble ML model - R^2 0.98125, >90% predictive accuracy	Proactive waste planning, 8 waste/valorization outputs predicted
Waste Valorization Planning	No secondary product estimation	Potash, Epsom Salt, Magnesium Oil predicted automatically	Revenue opportunities from waste by-products identified before harvest
Cross-site Learning	No data sharing possible - privacy/competition barriers	Federated Learning - 0 bytes raw data shared, collaborative model improvement	Privacy-preserving multi-site learning
ML Runtime	Python + TensorFlow (IPC overhead)	TypeScript + ONNX Runtime (native NestJS)	Single runtime stack, eliminated cross-language overhead

3. Quantitative Evidence

Crystal

- LSTM Validation $R^2 = 0.8884$ - model explains ~89% of variance in 8 crystallization parameters
- Test MAE = 0.2264, Test RMSE = 0.3651 on unseen data
- Input: 30-day window $\times$ 15 features | Output: 60 days $\times$ 8 parameters = 480 values per inference
- Linear Regression chosen over ARIMA for monthly production calibration
- $\pm 15\%$ confidence bounds on monthly production forecasts
- Forecasts scaled landowner-wise using individual bed count ratio against 7,500 total facility beds

Compass

- Demand forecast uses seasonal yield ratio algorithm - Yala: 8.74%, Maha: 10.27% - derived from 168 months of PSS historical data (2012–2025)

- Demand validation error: ~10% MAPE on held-out PSS historical data
- Demand per-landowner - scaled by individual bed count ratio, linked directly to Crystal production forecast
- SARIMAX price model MAPE: 1.16% month+1, 1.35% month+2 - validated using rolling-origin cross-validation
- SARIMAX price MASE < 1 - confirmed beating naive baseline
- Price model runtime inputs: season flag, price lag 12 months, latest actual price, PSS production volume
- Two-pass recursive forecasting for month+2 - month+1 prediction used as input to month+2

Vision

- Average processing time: ~45ms/frame (confirmed 40–50ms range in test data)
- Purity calculation verified: 71.43% and 75.00% → average 73.22%
- 24 hourly bins for temporal analytics
- 3-class NMS deduplication at IoU 0.45
- 320×320 input enabling ~20+ FPS on CPU

Valor

- Overall $R^2 = 0.98125$ - ensemble model explains ~98% of variance in waste data
- Per-target R^2 range: 0.9676 – 0.9907 across all 8 waste categories
- Total_Waste_kg $R^2 = 0.9907$ - highest performing target
- Mean inference time: ~285ms - sub-second warm prediction
- Improvement over manual estimation: ~60–65% (manual) → >90% (BrineX-Valor)
- 8 output targets predicted per inference: 4 solid, 1 liquid, 3 valorization products

System-Wide

- 11 microservices (9 gRPC + 2 Kafka consumers)
- 11 proto files - type-safe gRPC contracts
- 51 automated test spec files across all services
- 10 CI/CD workflows (path-filtered GitHub Actions)
- 8 user roles across 3 subscription tiers
- 16+ MongoDB collections across all services

4. Qualitative Evidence

Crystal

- LSTM captures both short-term daily fluctuations and long-term seasonal patterns via 30-day sliding window
- Linear Regression selected over ARIMA after comparative testing - simpler, more stable for calibration task
- Weather integration (7 OpenWeatherMap features) accounts for environmental variability in crystallization rates
- Landowner-wise scaling enables individual production planning from a single facility-level forecast

- Multi-horizon output supports both operational (daily) and strategic (seasonal) planning

Compass

- Demand forecast uses yield ratio algorithm - simple, interpretable, and directly linked to Crystal production forecasts - landowner's own sensor readings ultimately drive their demand estimate
- Seasonal yield ratios (Yala: 8.74%, Maha: 10.27%) derived from 168 months of PSS aggregate data - captures Maha/Yala buying behavior differences (distributors book more aggressively in Maha due to quality premium)
- Farm-specific ratios will automatically replace regional ratios after 6–12 months of individual deal history - no code changes required
- SARIMAX chosen for price prediction - 168 months of monthly price data is ideal scale for this model class, captures seasonal price structure (Maha/Yala) and supply volume sensitivity natively
- SARIMAX price model MAPE 1.16% confirms strong price level accuracy - directional accuracy (46.3%) is low but operationally irrelevant since landowners need price level estimates, not direction
- Seller transparency - ranked by price per bag and bag count, criteria fully visible to landowners, no hidden algorithm bias

Vision

- 3-layer guard stack ensures only Laboratory plan users access Vision module
- WebSocket real-time architecture delivers live detection results with sub-second latency
- HSV-based whiteness scoring provides pixel-level calibratable measurement independent of human judgment
- Batch management with timestamps enables full quality traceability per production run
- ROI-based filtering allows targeted analysis of specific camera regions

Valor

- Federated Learning architecture - FedAvg for neural network components, quality-weighted model selection for tree-based models
- Client-side local training - 0 bytes of raw operational data shared with central server
- Supports 5–10 nodes (prototype), 10–30 nodes (pilot), 100+ nodes (production)
- Ensemble model integrates: XGBoost/Gradient Boosting, stacked tree ensembles, deep neural network, Ridge regression meta-learner
- Event-driven prediction workflows via AWS SQS - asynchronous job processing
- Feature engineering pipeline: log/scaled transformations, seasonal sin/cos encoding, brine density proxy indicators

System-Wide

- Separation of concerns: each microservice owns its data domain
- Type safety: gRPC proto contracts across all 11 services
- Event-driven decoupling: Kafka for email and audit - zero blocking on user requests

- Defense in depth: JWT → Roles → Plan guard stack at API Gateway
- i18n: Sinhala, Tamil, English via next-intl

5. Data-Backed Claims and Justified Conclusions

Claim	Evidence	Justification
LSTM achieves 88.84% validation accuracy on daily parameter prediction	$R^2 = 0.8884$, Test $R^2 = 0.7750$, MAE = 0.2264	~89% variance explained - sufficient for production planning with $\pm 15\%$ confidence
Linear Regression outperforms ARIMA for monthly production calibration	ARIMA tried and rejected - Linear Regression selected for better seasonal fit	Simpler model more appropriate for calibration from LSTM output
Monthly production forecasts delivered landowner-wise with $\pm 15\%$ confidence	Calibration constants applied per landowner bed count ratio against 7,500 total beds	Individual production planning enabled from facility-level forecast
Demand forecast achieves ~10% MAPE on held-out PSS data	Validated against held-out PSS historical data — 168 months (2012–2025)	Seasonal yield ratios (Yala: 8.74%, Maha: 10.27%) derived from regional aggregate data - sufficient accuracy for workforce allocation and distributor scheduling
SARIMAX price model achieves 1.16% MAPE month+1, 1.35% month+2	Rolling-origin cross-validation — leakage-free, each month forecast using only prior data	Price level accuracy is operationally relevant - directional accuracy (46.3%) is low but expected due to mean-reverting market behavior
Real-time salt detection at ~45ms/frame	performance.now() confirmed 40–50ms range	320×320 ONNX with graphOptimizationLevel: 'all' enables ~20+ FPS on CPU
Objective quality measurement replaces subjective inspection	HSV formula $V \times (1-S)$ per pixel - repeatable, calibratable	Pixel-level analysis independent of human judgment
Valor ensemble explains ~98% of variance in waste data	Overall $R^2 = 0.98125$, per-target R^2 0.9676–0.9907 on held-out test data	Strong predictive consistency across all 8 waste and valorization categories
Valor improves waste estimation from ~60% to >90% accuracy	Manual estimation ~60–65% vs ML model >90% - internal testing	Proactive waste management enabled replacing reactive manual estimation
Federated Learning enables cross-site model improvement without data sharing	FedAvg aggregation - client-side local training only, 0 bytes raw data transmitted	Privacy-preserving collaborative learning across competing salt production sites

Sub-second waste prediction inference	Mean ~285ms across 50 benchmark runs	Warm inference suitable for real-time production decision support
Full audit trail with zero latency impact	Kafka fire-and-forget - 6 MongoDB compound indexes	Non-blocking async logging, 16 fields per event
Microservices enable independent scaling	11 services - independent Dockerfiles, CI pipelines, port assignments	Vision scales for inference spikes, Auth scales for OTP spikes independently